Lab 6 | Blackjack UML Class Diagram

Updated to reflect the Python classes and methods discussed for Tasks 1–10

card

+ rank
+ suit

+ __init__(rank, suit)
+ __str__()

deck

+ cards
+ needs_shuffle

+ __init__(num_decks=6)
+ shuffle()
+ draw()
+ add_plastic_card()

player

+ name
+ chips
+ hand
+ bet

+ __init__(name, chips=100)
+ hit(deck)
+ stay()
+ place_bet(amount)

game

+ deck
+ players
+ dealer

+ __init__(num_players=1)
+ play_round()

hand

+ cards

+ __init__()
+ add_card(new_card)
+ get_value()

Relationships:

- deck contains card objects
- hand contains card objects
- player has a hand
- dealer inherits player
- computer_player inherits player
- counting_player inherits player
- game uses deck, players and dealer

dealer

+ (inherits player)

+ __init__()
+ take_turn(deck)

computer_player

+ (inherits player)

+ take_turn(deck)

counting_player

+ count
+ threshold

+ __init__(name, chips, threshold)
+ count_card(c)
+ take_turn(deck)

+ indicates a public attribute or method. Review game class members against your saved final notebook.